DiaTool-DPO

Enhancing Tool-Augmented LLM Dialogue via Direct Preference Optimization

Automatic paired trajectory construction for improving TA-LLM slot-filling and tool call rejection — without human labeling

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Tool-Augmented LLMs must decide whether to ask follow-up questions, make a tool call, or reject inappropriate requests. SFT with expert trajectories alone has critical limitations.

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Missing Follow-ups

Model skips asking for required parameters, proceeding with incomplete information

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Hallucinated Tool Calls

Model fabricates tool invocations with incorrect or imagined arguments

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Failed Rejections

Model attempts tool calls for out-of-scope requests instead of declining

DiaTool-DPO proposes Direct Preference Optimization to enhance dialogue capabilities by automatically constructing paired trajectory datasets of correct and incorrect dialogue flows.

Modeling TA-LLM Dialogue as a 5-State MDP

S1
Initial
No history; reject out-of-scope

S2
Tool Selected w/o Slots
Tool identified; params missing

S3
Tool Selected w/ Slots
All arguments determined

S4
Wait for Response
Tool call executed

S5
Complete
Results received; completion sent

Key Transitions

S1 → S3 → S4 → S5	Complete query — all required arguments present, direct tool call
S1 → (S2×N) → S3 → S4 → S5	Incomplete query — slot-filling QA repeats N times to gather missing parameters
S1 → S1	Out-of-scope query — no suitable tool available, request rejected

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Scenario	Example
Type 1 **Sufficient Info**	1 → 3 → 4 → 5	Query contains all required arguments. No slot-filling needed.	"Book a flight from NYC to LA on Dec 25" — all parameters present, direct tool call
Type 2 **Incomplete Info**	1 → (2×N) → 3 → 4 → 5	Query lacks some arguments. Slot-filling QA repeats N times until all fields gathered.	"Book a flight to LA" — missing date & origin, model asks follow-up questions
Type 3 **Out-of-Scope**	1 → 1	Requested functionality is unavailable. Tool call must be rejected.	"What's the meaning of life?" — no suitable tool available, model returns rejection

KEY INSIGHT

This 3-type classification, combined with the 5-state MDP, enables systematic construction of preference pairs for DPO training — each query type defines which trajectories are correct vs. incorrect.

Automatic Paired Trajectory Construction Without Human Labels

Query	Chosen Trajectory	Rejected Trajectory	Learning Lesson	Count	Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2×N)→3→4→5	Prevent redundant slot-filling	562 / 2,530	Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,090 / 562	Easy/Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	Easy
Type 2	1→(2×N)→3→4→5	1→3→4→5 / 1→(2×M)	Prevent slot hallucination	562 / 2,530	Hard
Type 3	1→1	1→3→4 / 1→(2×N)→3→4	Tool call reject	567 / 562	Hard

Automatic Construction

No human labeling required. Pairs are generated from the MDP structure and query type classification.

Learning Signal

Each pair teaches a specific lesson: prevent redundant questions, prevent hallucination, or enforce rejection of unavailable tools.

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

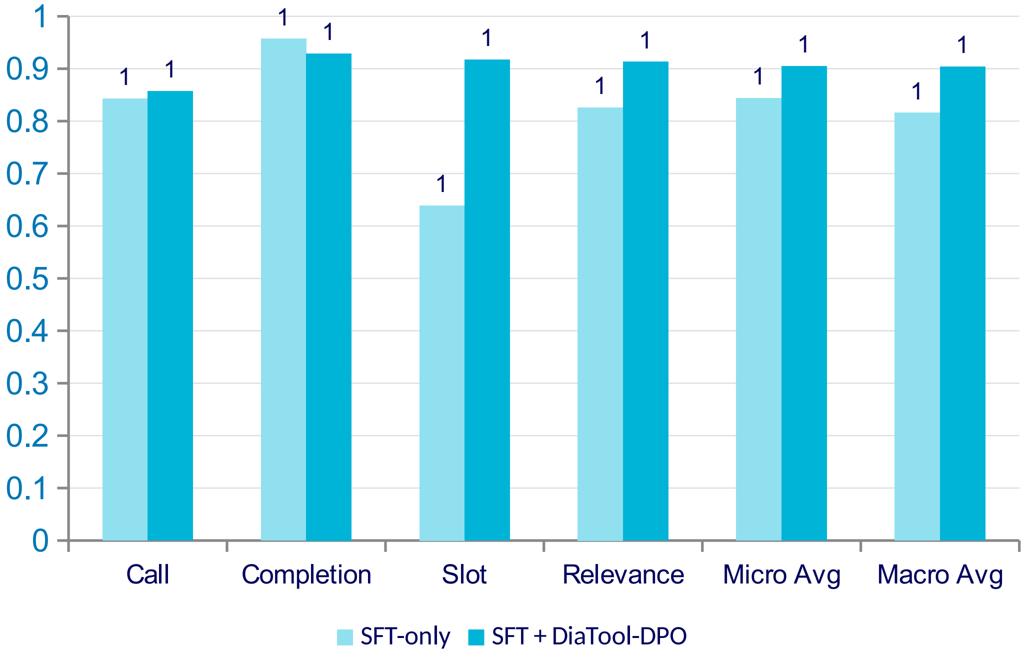
FunctionChat-Bench Results

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	**DPO-only**	0.314	0.700	**0.833**	0.609	0.575	0.614
	SFT-only	**0.900**	0.916	0.694	**0.913**	0.870	0.856
	SFT+DPO	0.886	**0.929**	**0.833**	0.826	**0.884**	**0.868**
Prop.-3.1B	**DPO-only**	0.357	0.551	0.528	0.391	0.455	0.457
	SFT-only	**0.771**	0.817	0.750	**0.826**	**0.790**	0.791
	SFT+DPO	0.743	**0.871**	**0.833**	**0.826**	0.765	**0.818**
LLaMA-3-8B	**DPO-only**	0.029	0.449	0.056	0.261	0.205	0.199
	SFT-only	0.843	**0.957**	0.639	0.826	0.844	0.816
	SFT+DPO	**0.857**	0.929	**0.917**	**0.913**	**0.905**	**0.904**

SFT + DiaTool-DPO consistently achieves the highest Macro Average across all model sizes, demonstrating that DPO complements SFT for balanced dialogue control performance.

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance

Impact of adding DiaTool-DPO to SFT training on LLaMA-3-8B



Slot Accuracy

+43.5%

0.639 → 0.917

Relevance (Tool Rejection)

+10.5%

0.826 → 0.913

Near GPT-4o performance: 94.8% information gathering, 91% tool call rejection

Key Takeaways

RL-Based Dialogue Control for Production TA-LLMs

01

MDP Formulation Enables Systematic Pairing

Defining 5 internal states and 3 query types allows automatic generation of chosen/rejected trajectory pairs without human annotation.

02

DPO Complements SFT, Not Replaces It

DiaTool-DPO alone performs poorly. Combined with SFT, it consistently achieves the best balanced Macro Average across all model sizes.

03

Largest Gains in Slot-Filling and Tool Rejection

LLaMA-3-8B improves Slot accuracy by +43.5% and Relevance by +10.5% over SFT-only, reaching near GPT-4o performance.

04

Language-Agnostic and Production-Ready

While primary experiments were in Korean, the method generalizes to English — opening broad applicability for production TA-LLMs.